Figure 1: Research Framework Architecture — Flowchart showing the complete ten-stage pipeline of the proposed framework: Sensor Data Acquisition, Data Preprocessing, Feature Engineering, the XGBoost Predictive Model, SHAP Explainability, Dynamic Threshold-Based Risk Assessment, Digital Twin Monitoring and Visualisation, the Repair vs. Replace Decision Module, Maintenance Recommendation, and the Feedback Loop for Continuous Learning. Arrows indicate the sequential flow of information from raw sensor input through to an explainable, cost-aware maintenance recommendation; the dashed return arrow represents the continuous learning loop, whereby maintenance outcomes are fed back to retrain the model and recalibrate the intervention threshold. Source: Author's work.

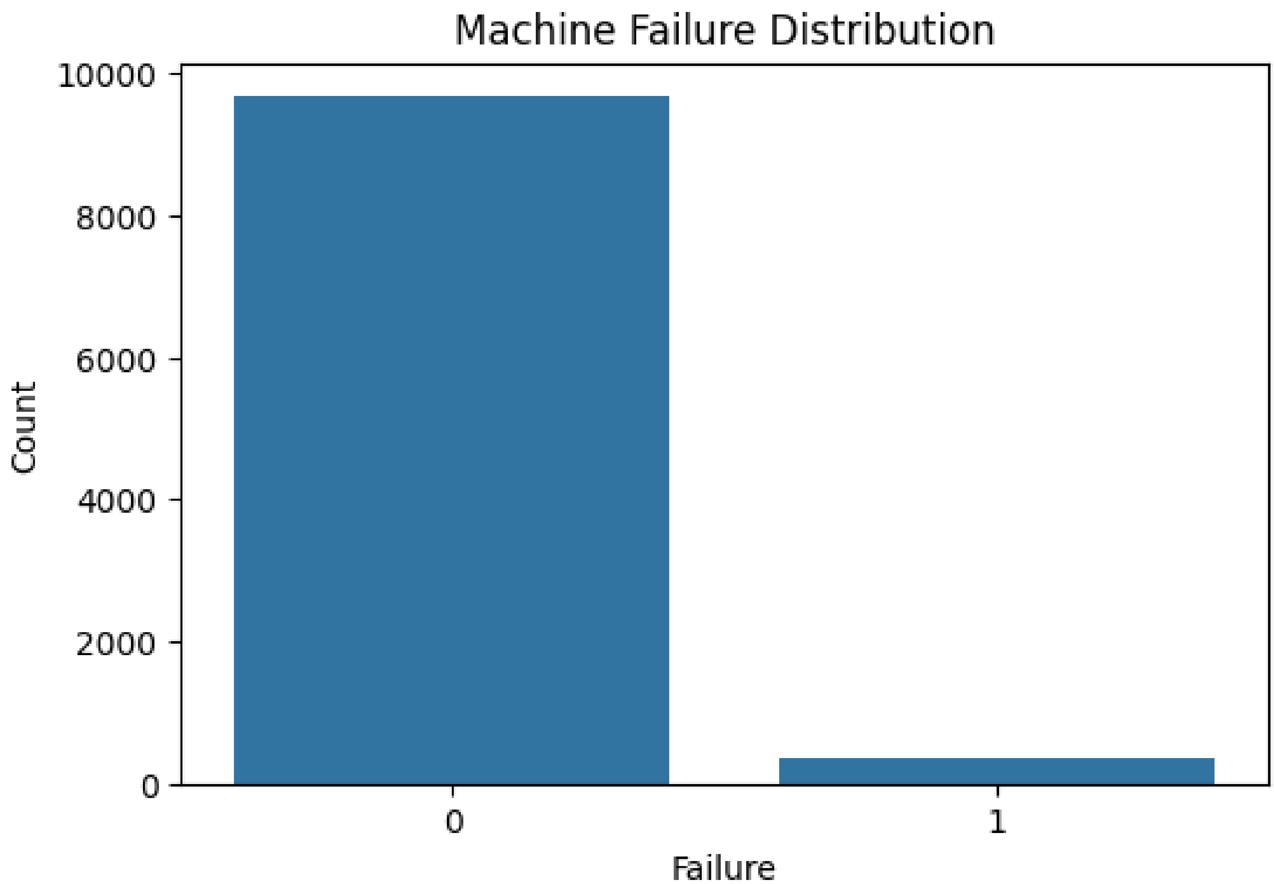

Figure 2: Machine Failure Distribution — Bar chart showing 9,661 non-failure observations versus 339 failure observations. Source: Author's work.

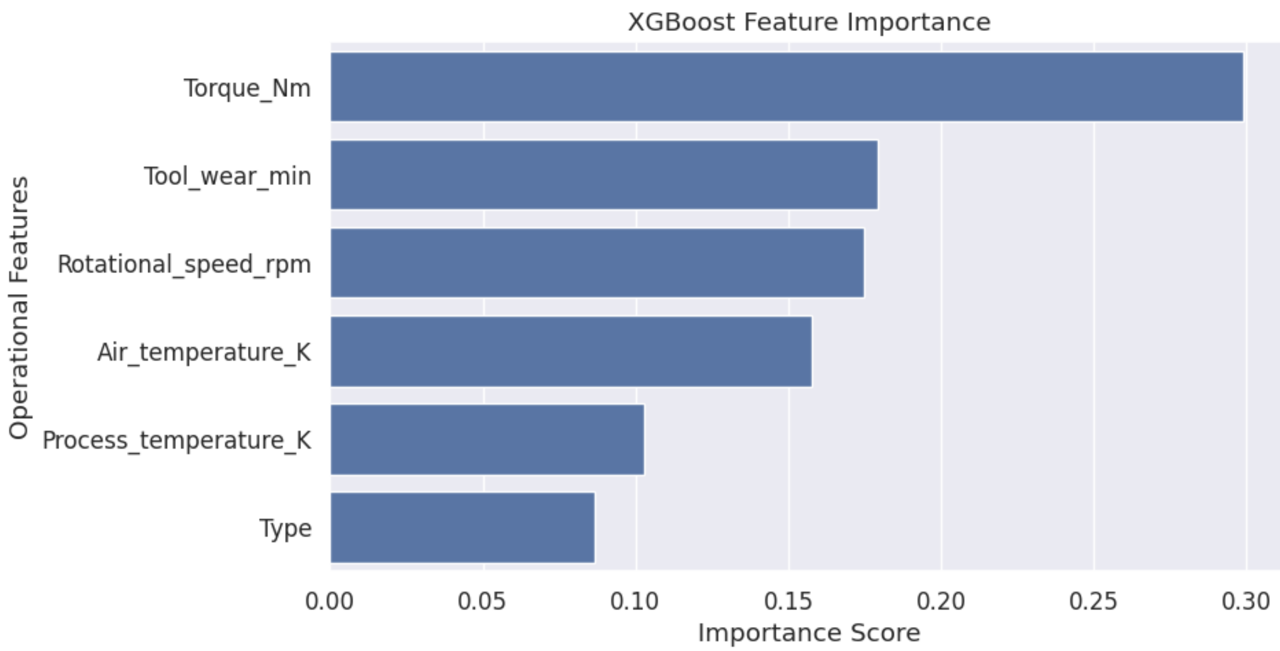

Figure 11: Feature Importance Ranking — Horizontal bar chart of all six features ranked by XGBoost importance score. Torque highest at 0.299, Machine type lowest at 0.087. Source: Author's work.

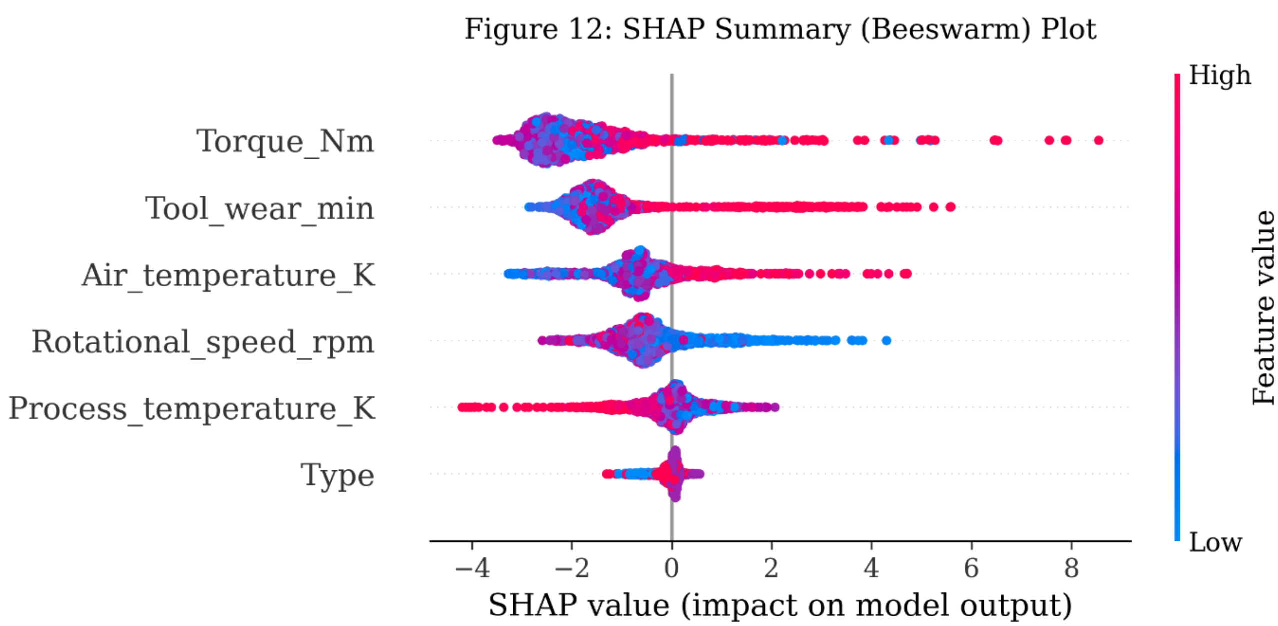

Figure 12: SHAP Summary (Beeswarm) Plot — Global distribution of SHAP values for all six operational features across the 2,000-observation test set. Each point represents one machine observation. Horizontal position indicates the magnitude and direction of the SHAP contribution to failure probability; colour encodes the raw feature value (red = high, blue = low). Features are ranked vertically in descending order of mean absolute SHAP value. Generated using shap.plots.beeswarm() applied to the TreeSHAP explainer fitted on the trained XGBoost classifier. Source: Author's work.

Figure 14: SHAP Waterfall Plot (High-Risk Machine)

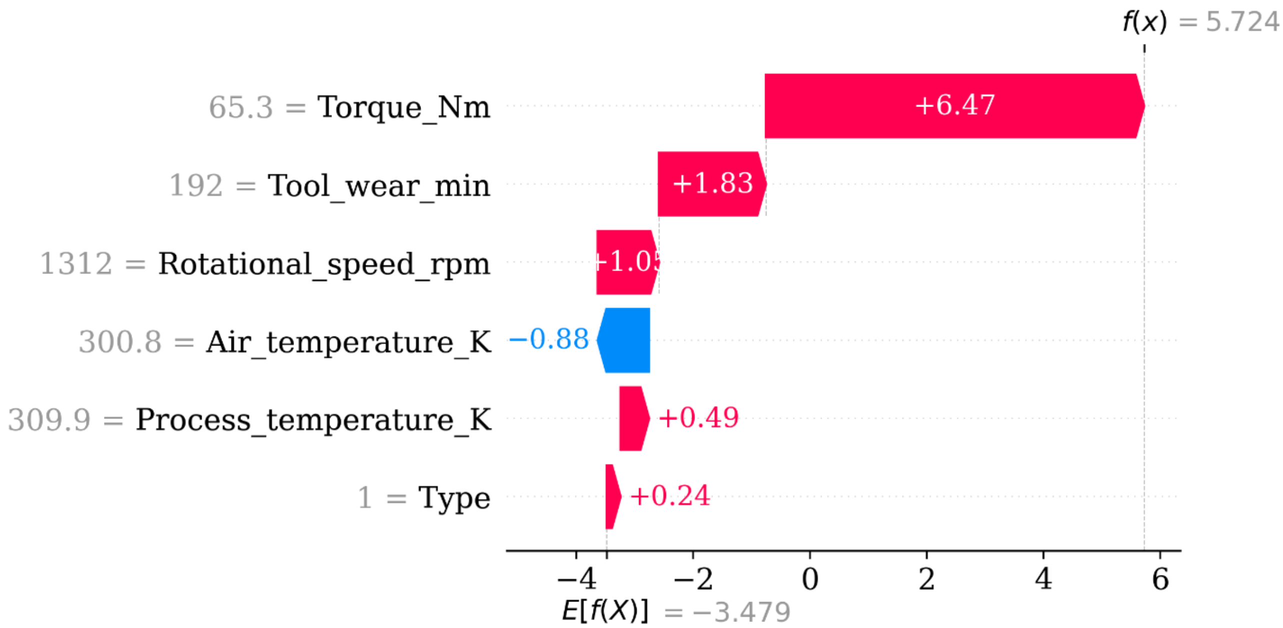

Figure 14: SHAP Waterfall Plot — Instance-level SHAP decomposition for a representative High Risk machine (P(Failure) ≥ 0.70) from the test set. The plot begins at the base value E[f(x)] (mean XGBoost failure probability across the training population) and shows the additive contribution of each feature, terminating at the individual machine's predicted failure probability f(x). Red bars indicate features that increase the failure probability prediction; blue bars indicate features that decrease it. Generated using shap.plots.waterfall() applied to a single test-set observation. Source: Author's work.

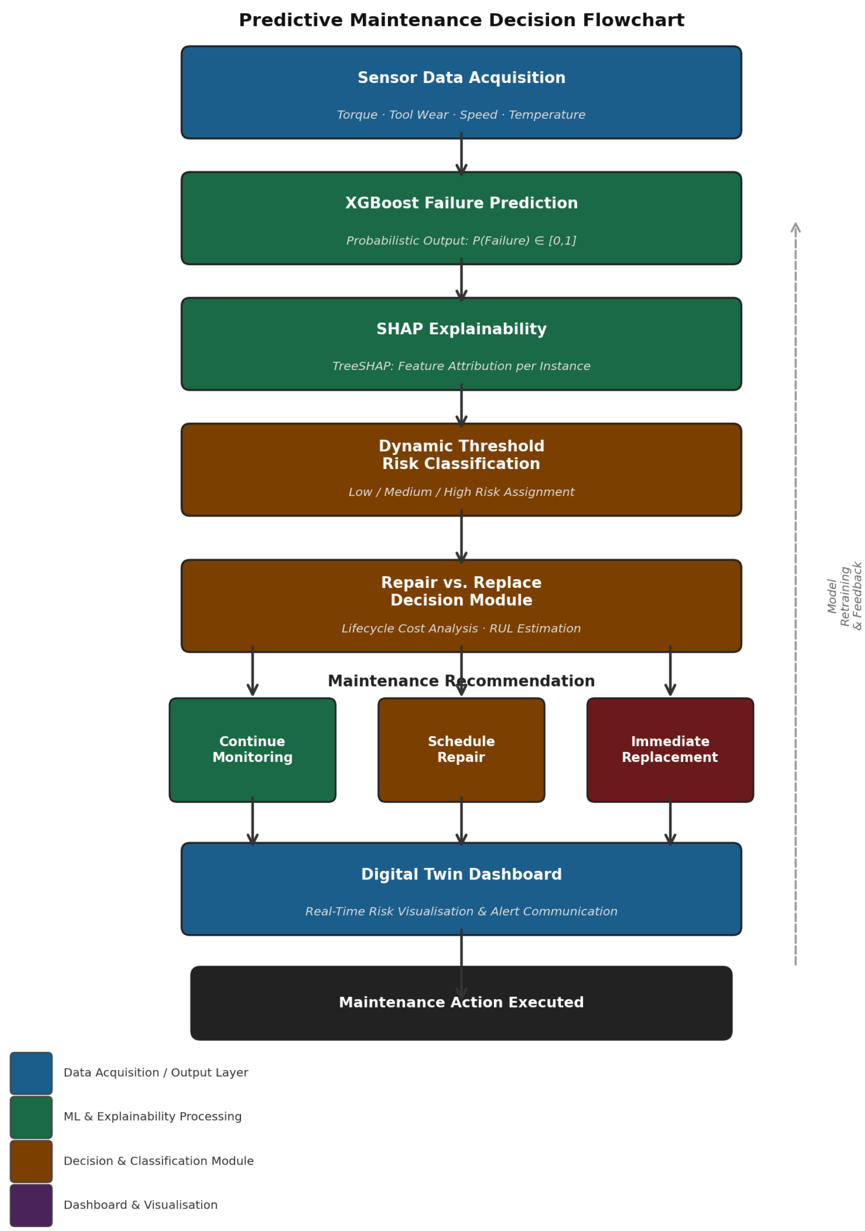

Figure 17: End-to-End Predictive Maintenance Decision Flowchart — Illustrating the complete operational pipeline from Sensor Data Acquisition through XGBoost Failure Prediction, SHAP Explainability, Dynamic Threshold Risk Categorisation, Repair vs. Replace Decision Module, Maintenance Recommendation (Continue Monitoring / Schedule Repair / Immediate Replacement), and Digital Twin Dashboard integration. The dashed feedback loop on the right represents continuous model retraining and threshold calibration from maintenance outcome records. Source: Author's work.